

Article

A Candidate-Free Location Optimization Framework for Gas Repair Stations Under Stochastic Road Resistance Conditions

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Abstract

Emergency response in urban gas pipeline networks is highly sensitive to stochastic traffic conditions, which introduce substantial uncertainty in crew travel times to leakage sites. Existing facility location models typically rely on predefined candidate sites and deterministic travel assumptions, limiting their ability to capture full-cycle dynamic recovery processes under random leakage events and traffic congestion. This study develops a candidate-free location optimization framework for repair station siting under stochastic road resistance conditions, aiming to characterize spatial variability in emergency response capability. The framework integrates a candidate-free facility location model with a hybrid greedy–Monte Carlo solution strategy to optimize station layouts across network-wide stochastic scenarios. Coverage reliability, response time, and construction cost are jointly considered to support robust siting decisions. A case study based on the real road and gas pipeline networks of City H demonstrates the effectiveness of the proposed approach. Across 20,000 stochastic road resistance scenarios, the optimized layout achieves an average service coverage rate of 97.77% within the specified response time threshold, while maintaining stable performance under variability. Although increasing the number of stations enhances response capability, the improvement exhibits clear diminishing marginal returns. These findings provide quantitative guidance for determining cost-effective station scale and prioritizing core hub locations under uncertainty. The proposed framework offers a structured decision-support tool for resilience-oriented planning, prioritization of critical segments, and evaluation of emergency response and maintenance strategies in urban gas pipeline systems.



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Keywords: gas emergency repair station; candidate-free location; stochastic road resistance; improved greedy algorithm; Monte Carlo simulation

1. Introduction

As a critical component of urban lifeline infrastructure, the safe and stable operation of gas pipeline networks is directly linked to public safety, economic development, and social stability [1,2]. Emergency response capability determines whether localized failures escalate into systemic risks. Gas leakage incidents are highly destructive and prone to rapid escalation, making the establishment of an efficient and reliable emergency repair

system essential. Within such a system, the scientific siting of repair stations constitutes a core determinant of emergency response efficiency [3,4]. To mitigate the risks associated with pipeline failures, the establishment of dedicated gas emergency repair stations has become a fundamental and widespread physical safety measure in major cities around the world. These stations are responsible for rapidly dispatching emergency repair crews and equipment to the site of a leak; they are not merely auxiliary facilities, but rather a standard and indispensable component of modern urban gas safety management systems.

In the context of rapid urbanization, the operational environment of gas pipeline networks has become increasingly complex. In recent years, the national natural gas pipeline infrastructure has undergone large-scale expansion and upgrades. As of the end of 2024, the total length of long-distance natural gas pipelines nationwide had exceeded 120,000 km, with the main trunk network spanning over 57,000 km. While the rapid expansion and increasing complexity of the pipeline network have enhanced gas supply capacity, they have also significantly increased the system's exposure to operational risks and the challenges of emergency management. The Ministry of Housing and Urban–Rural Development has explicitly emphasized in the “14th Five-Year Plan for National Urban Infrastructure Development” the need to strengthen urban safety resilience and optimize the spatial allocation of emergency resources [5,6]. Meanwhile, frequent uncertainties such as peak-hour congestion and road construction introduce significant stochastic variability into the travel times of repair vehicles [7–9]. When pipeline failures occur, delays caused by suboptimal station layouts may not only trigger secondary disasters and considerable economic losses but also threaten social order and public safety [10,11]. For example, after a natural gas pipeline in Luzhou, Sichuan, was damaged during construction, delayed emergency response allowed the pipeline to operate under high pressure for approximately 13 h, resulting in direct economic losses of about 2.8 million yuan; in Shenyang, a gas pipeline was severed during construction, causing nearly 1000 households to be without gas for nearly 14 h. These realities highlight that scientifically optimized repair station siting is fundamental to enhancing the resilience of urban gas systems.

Research on emergency facility location has developed along two primary directions. One focuses on locating resource points for repairing and restoring damaged infrastructure following natural disasters or sudden incidents [12–15], while the other concerns siting emergency centers for delivering rescue supplies and personnel to accident sites [16–19]. Although these streams differ in operational focus, they share a common objective: maximizing service effectiveness under constrained resources. This study focuses on the first stream, namely the location optimization of gas emergency repair stations. Through scientifically designed spatial layouts, the repair network aims to respond rapidly to failures occurring anywhere within the pipeline system, despite the influence of stochastic road resistance, thereby strengthening the overall resilience of the gas supply network.

From a modeling perspective, the unpredictable spatial occurrence of pipeline failures naturally places this problem within the framework of covering location models [20,21]. However, traditional approaches in this domain typically rely on a predefined set of candidate sites. Such candidate sets are often determined by empirical judgment or simplified rules and may fail to include globally optimal potential locations. Furthermore, under stochastic road resistance, the adaptability and robustness of candidate-based solutions are often limited, thereby constraining the optimality of the feasible solution space [22–24]. Classic models such as the Maximum Coverage Location Problem (MCLP) [25–27] and the Location Set Covering Problem (LSCP) [25,28,29] are predominantly formulated under deterministic environmental assumptions, making them less capable of capturing the time-varying stochastic characteristics of real-world traffic. Although subsequent extensions, such as the Gradual Coverage Location Problem (GCLP) [21,30,31],

have improved modeling flexibility, challenges remain in achieving sufficient optimization granularity and robustness when confronting complex network structures combined with environmental uncertainty.

In response to these limitations, this study investigates a Candidate-Free Stochastic Road resistance Location Problem (CF-SRLP). The central research question is: under stochastic road resistance, how can one search directly across the entire network domain, without presetting candidate points, for a station layout that maximizes expected coverage reliability? The differences between the model proposed in this paper and the traditional model are shown in Table 1.

Table 1. The differences between the proposed model and traditional models.

Paper	Model Objective	Uncertainty Factors	Model
Arslan O. (2021) [25]	Maximize coverage given a fixed number of facility locations	site costs, site coverage	LSCP, MCLP
Muren et al. (2020) [26]	Given a fixed number of facility locations, maximize coverage while minimizing service level imbalances among the facilities	site costs, site coverage, workload	BMCLP
Erdemir E T (2010) [28]	Maximize coverage given a fixed number of facility locations	Station costs, station coverage, travel time (excluding traffic delays)	LSCP, MCLP
Simoyama F O (2025) [31]	Given the limited number of facilities that can be built, maximize the coverage of all areas to their desired levels, giving priority to areas with higher weightings.	Site cost, site coverage, distance attenuation function	GCLP
This Paper	Under conditions of random road resistance, the facility layout that maximizes expected coverage reliability is identified directly across the entire network domain without the need to predefine candidate points.	site costs, station coverage, travel time (accounting for time-varying road conditions)	CF-SRLP

This study makes three primary contributions. First, it overcomes the limitations of traditional candidate preselection by proposing a candidate-free search paradigm across the entire network domain. Second, it develops an optimization model that explicitly incorporates stochastic road resistance, thereby aligning more closely with real-world operational conditions. Third, it designs a hybrid solution strategy that combines an improved greedy algorithm with Monte Carlo simulation to efficiently generate a Pareto-optimal solution set. A case study demonstrates the effectiveness and practical value of the proposed model and algorithm. Overall, this research aims to provide both theoretical foundations and practical decision-support tools for urban gas safety management authorities.

The remainder of this paper is organized as follows. Section 2 describes the repair station location optimization problem, including model assumptions and the methodological framework. Section 3 presents the mathematical model formulation and the solution algorithm. Section 4 reports a case study based on the topological network structure of City H, deriving siting schemes and conducting comparative feasibility analyses. Section 5 concludes the study and outlines directions for future research.

2. Problem Formulation and Conceptual Framework

In this section, we first define the repair station location problem and the relevant emergency response modes. We then introduce the modeling assumptions and finally present the overall methodological framework, which integrates network construction, stochastic travel time modeling, and optimization.

2.1. Problem Definition and Emergency Response Modes

The core of the gas emergency repair station location optimization problem lies in scientifically determining the spatial layout of repair stations based on the structural characteristics of the existing urban road network under limited construction resource constraints. Specifically, the task is to select an optimal combination of stations from all possible road node locations so that repair vehicles can cover as many demand nodes as possible within a stipulated response time threshold, thereby maximizing emergency service coverage.

At present, response modes for urban emergency rescue facilities can generally be classified into three typical types [32]. The first is the zone-based response mode, in which each demand point is assigned to its jurisdictional rescue station according to predefined administrative boundaries, as illustrated in Figure 1a. The second is the travel-time-based response mode, where each demand point is connected to the repair station with the shortest travel time, as shown in Figure 1b. The third is the Euclidean-distance-based response mode, which assumes ideal traffic conditions and assigns each demand point to the geographically nearest station, as presented in Figure 1c.

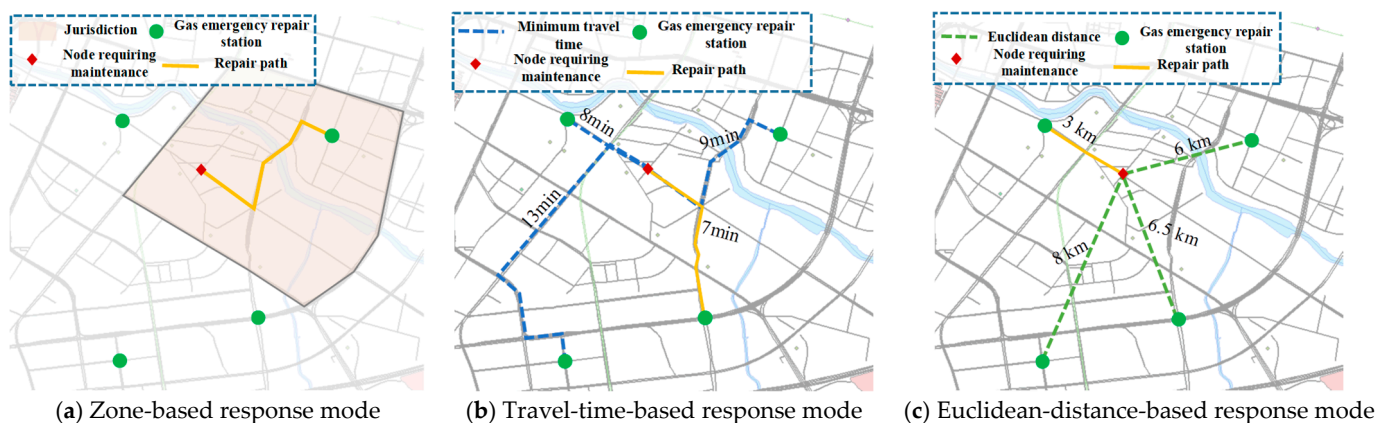


Figure 1. Schematic diagram of the three emergency response modes.

According to the Technical Standard for Safety Operation Monitoring of Urban Lifeline Engineering [33], risk warning for urban lifeline projects requires standardized emergency response procedures, mandating that repair teams arrive at accident sites within a prescribed response time threshold. As core facilities for emergency resource reserve and personnel dispatch, gas repair stations directly determine emergency response speed, network coverage capability, and overall repair efficiency.

In large cities or regions with complex road networks, the time-varying nature of traffic conditions significantly affects vehicle travel time. Therefore, the location of gas repair stations must conform to the travel-time-based response mode. The essence of this mode is to achieve maximum coverage of demand nodes under limited station resources by explicitly quantifying the dynamic characteristics of travel time under different stochastic road resistance scenarios.

However, mainstream candidate-based location models typically rely on a predefined set of fixed candidate points, which presents several inherent limitations in the context

of stochastic and dynamic urban environments. First, the preselection of candidates often relies on empirical judgment or simplified rules such as administrative boundaries, locations of existing facilities, land availability, or cost-based filters which may not be derived from a systematic optimization of coverage potential. Consequently, this predefined set may exclude spatially superior locations from the outset, imposing an upper bound on the final solution regardless of the sophistication of the subsequent location algorithm. Second, while advanced stochastic or robust optimization techniques can be applied within given candidate set to handle travel time uncertainty, the adaptability of the resulting layout is fundamentally constrained by that static spatial domain. A candidate set defined a priori cannot dynamically evolve with real-time or scenario-varying network conditions, potentially leading to coverage degradation when actual travel patterns deviate from the assumptions used during candidate generation. Third, and most fundamentally, candidate-dependent approaches inherently restrict the decision space to a discrete subset of the network. This restriction risks overlooking globally superior configurations that lie outside the predefined set, thereby limiting the model's ability to discover the most robust and efficient layout for the entire network under uncertainty. The proposed candidate-free paradigm aims to transcend these limitations by expanding the search to all network nodes, thereby directly coupling global spatial optimization with stochastic performance evaluation.

To address these issues, the candidate-free location method offers significant advantages. Rather than pre-specifying candidate sites, it expands the decision space to include all nodes in the road network and directly identifies the optimal station combination through optimization algorithms. First, it eliminates the constraint imposed by candidate preselection, thereby enabling a global search across the entire network and enhancing overall coverage potential. Second, it better reflects dynamic road environments, particularly under stochastic road resistance conditions, allowing a more realistic representation of travel time uncertainty and strengthening the robustness of the resulting layout. Third, it provides greater modeling flexibility, making it possible to integrate stochastic programming or chance-constrained approaches in order to maximize coverage effectiveness or minimize risk across multiple scenarios.

Accordingly, the fundamental idea of candidate-free location is to treat the complete set of road network nodes as the decision space, construct an optimization model aimed at maximizing demand coverage under a limited number of stations, and incorporate either a probability distribution or a scenario set of travel times under stochastic road resistance. By integrating stochastic simulation and heuristic search algorithms, the optimal station layout can be identified within the global solution space, ensuring high emergency response reliability even under variable traffic conditions.

In summary, this study establishes a stochastic road resistance-based candidate-free location optimization method for gas emergency repair stations to overcome the limitations of traditional candidate-based models. The approach achieves global optimization of repair station layout and enhances coverage effectiveness in dynamic road network environments, thereby providing a scientific foundation for urban gas safety emergency system planning.

2.2. Model Assumptions

When a gas pipeline accident occurs, a repair demand is generated, requiring a repair team to be dispatched from a repair station to the site within a specified time limit. According to the Code for Gas Engineering Projects [34], gas pipeline alignment must consider urban and rural roads as well as topographical conditions, providing a practical basis for this study.

However, due to the complexity of underground utility systems and variations across cities, gas pipelines do not always strictly follow road alignments. Their spatial layout may exhibit discontinuous and irregular patterns. To facilitate modeling and analysis, this study approximates the coupling relationship between the pipeline network and the road network using the urban road network topology. Pipeline repair demand points are simplified as their geographically nearest road nodes, as illustrated in Figure 2. It should be noted that this simplification may introduce some error in estimating actual response times. In practical applications, a more precise coupled pipeline–road network topology may be adopted for further refinement.

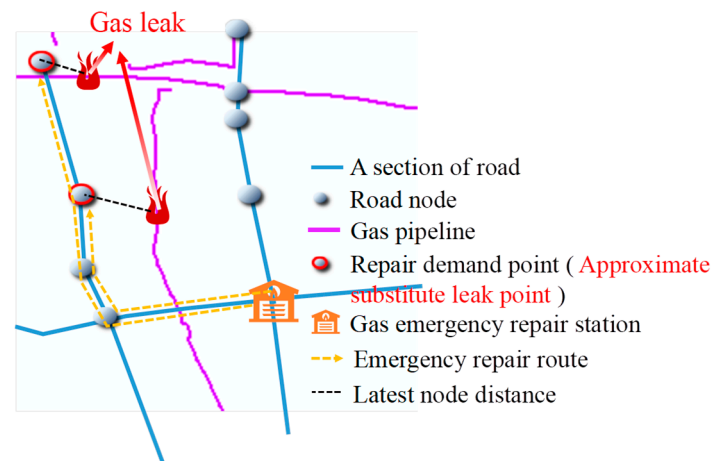


Figure 2. Schematic diagram of repair demand point setting.

To construct a tractable mathematical model while preserving the essential characteristics of the real system, the following assumptions are made [35,36]:

1. The coverage relationship depends solely on the shortest-path travel time. Queuing effects, task conflicts, and repair capacity limitations are not considered.
2. Traffic conditions are represented by stochastic road resistance coefficients. Each scenario is independent, and travel times vary across scenarios. Coverage relationships are evaluated separately under each scenario without inter-scenario interactions.
3. Candidate locations for repair stations are restricted to nodes in the road network topology. Deployment at non-node locations is not considered.
4. Construction cost is primarily determined by the number of stations. Differences in operational costs among stations are assumed to be negligible. This simplification allows the model to focus on the spatial efficiency of the layout under uncertainty. In practice, site-specific costs (e.g., land acquisition, construction standards) vary; the model's output of minimal station sets provides a Pareto-efficient frontier for coverage, upon which detailed cost–benefit analysis with heterogeneous costs can be conducted.
5. Under any given scenario, each demand point is served by the single repair station with the shortest travel time. Simultaneous multi-station response is not considered.
6. The shortest travel time between any two nodes satisfies symmetry, implying that the road network is bidirectionally connected.
7. Differences in demand intensity among nodes (e.g., user count or priority level) are not considered.

2.3. Methodological Framework

This study constructs a systematic research framework consisting of three core modules: Data Modeling, Optimization Solution, and Visualization and Verification, as illustrated in Figure 3.

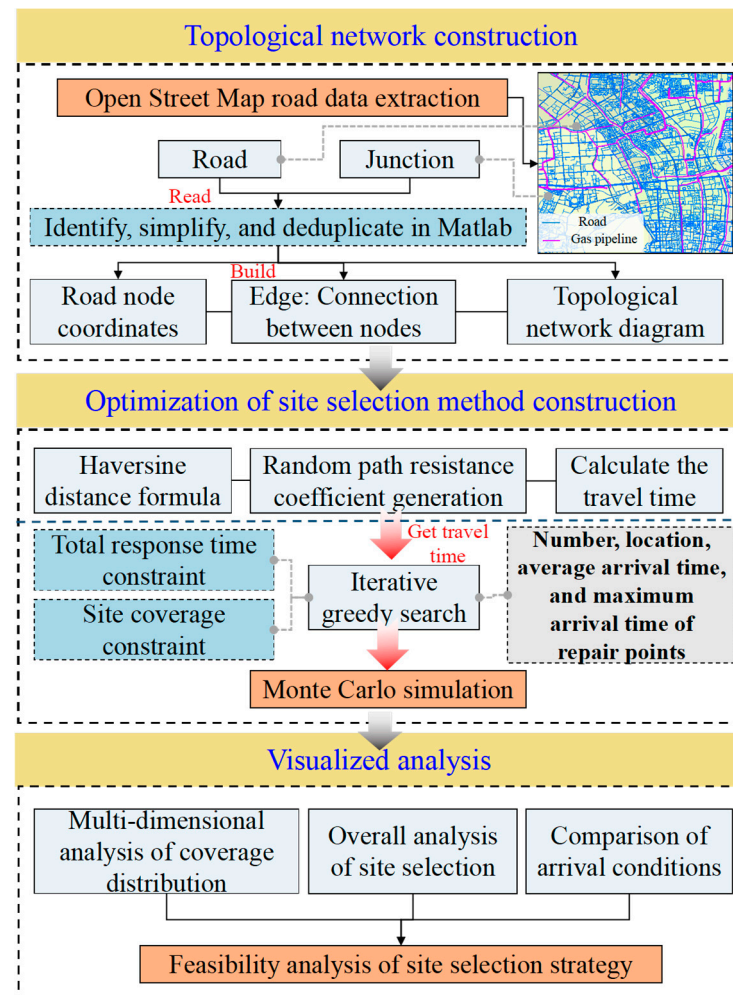


Figure 3. Schematic diagram of the research framework.

First, in the Topological Network Construction module, road network data are extracted from OpenStreetMap. Identification, simplification, and deduplication operations are performed in MATLAB 2023b, to generate a topological network structure containing node coordinates and connectivity relationships, providing the spatial foundation for subsequent modeling.

Second, in the Optimization of Site Selection Method Construction module, spatial distances between nodes are calculated using the Haversine formula. These distances are combined with randomly generated road resistance coefficients to derive dynamic travel times. Under constraints on response time thresholds and station coverage, an Improved Iterative Greedy Search algorithm is applied to optimize both the location and number of repair stations. Monte Carlo simulation is then conducted to obtain the distributions of coverage rate, average arrival time, and maximum arrival time across stochastic scenarios.

Finally, in the Visualized Analysis module, multi-dimensional statistical analysis is performed on the optimization results. This includes examination of station selection patterns, comparison of arrival-time performance, and evaluation of strategy feasibility.

Together, these components form a closed-loop analytical process from foundational data construction to optimized decision-making and performance verification.

3. Mathematical Model and Solution Algorithm

Gas incidents are characterized by sudden onset, high hazard potential, and stringent timeliness requirements, necessitating emergency response within the shortest possible time. To address the optimization problem of gas emergency repair station location, this study considers the randomness inherent in actual traffic conditions and establishes a candidate-free location optimization model based on stochastic road resistance.

This section presents the formal mathematical model and the corresponding solution procedure. It first describes road network topology construction, followed by stochastic travel time modeling, formulation of the objective function and constraints, and finally the improved iterative greedy search algorithm combined with Monte Carlo simulation.

3.1. Road Network Topological Construction

Based on the aforementioned road network processing requirements, it is necessary to extract the urban road network structure and the accessibility between road nodes from the original latitude and longitude information [35]. A schematic diagram of the road network processing is shown in Figure 4.

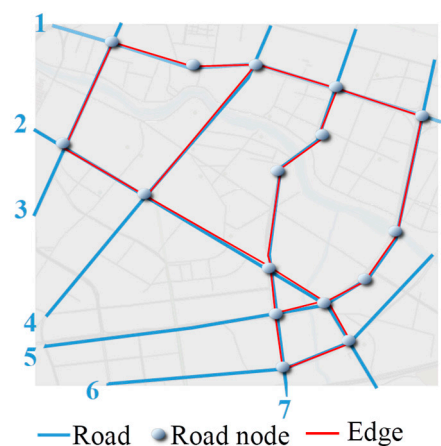


Figure 4. Schematic diagram of road network.

Figure 4 outlines the key steps for road network processing as follows:

1. The acquired raw road network data is imported into MATLAB for preprocessing. Each road geometry is segmented into several basic line segments, with each segment defined by the coordinates of its adjacent vertices.
2. All start and end points of the segments are included in the road node set, as shown in Equation (1):

$$V_{end} = \{(x_{i1}, y_{i1}) \cup (x_{i2}, y_{i2}) \mid \forall U_i\} \quad (1)$$

where V_{end} represents the set of all segment start and end points; U_i denotes the i -th road segment; (x_{i1}, y_{i1}) and (x_{i2}, y_{i2}) represent the coordinates of the start and end points of segment U_i , respectively.

3. If a key point exists on multiple road lines, it is considered an intersection point. The encapsulated polyxpoly function is used to detect valid intersections between line segments, as shown in Equation (2):

$$V_{int} = \{\text{polyxpoly}(W_j, W_k) \mid j, k \in \{1, \dots, N_{seg}\}, j < k\} \quad (2)$$

where V_{int} represents the set of all segment intersection points; j and k are indices of two segments, with $j < k$ indicating a no-repeat constraint; N_{seg} denotes the total number of segments after road segmentation.

4. V_{end} and V_{int} are merged. Geometric coincident points are eliminated through numerical tolerance filtering, and the latitude and longitude coordinates of all road nodes are recorded, as shown in Equation (3):

$$V_{nodes} = \text{uniquetol}(V_{end} \cup V_{int}, \varepsilon = 10^{-6}) = \{V_1, V_2, \dots, V_n\} \quad (3)$$

where uniquetol represents a tolerance-based unique function, and n denotes the number of recorded nodes. The tolerance threshold parameter ε addresses coordinate deviations during node generation. This paper sets $\varepsilon = 10^{-6}$, meaning points whose true distance is less than 10^{-6} m are considered the same point, while points farther apart remain independent.

Based on this, the road network is defined as a weighted undirected graph, as shown in Equation (4):

$$G = (V_{nodes}, E) \quad (4)$$

where $E = \{(V_a, V_b) | V_a, V_b \in V_{nodes}\}$.

3.2. Travel Time Modeling Under Stochastic Road Resistance

The Haversine formula accounts for Earth's curvature, making it more suitable than Euclidean distance for calculating distances between geographic coordinates, with precision meeting engineering requirements [37–39], as shown in Equation (5):

$$d_{ab} = 2R_{earth} \cdot \arcsin \sqrt{\sin^2\left(\frac{\varphi_b - \varphi_a}{2}\right) + \cos(\varphi_a)\cos(\varphi_b)\sin^2\left(\frac{\lambda_b - \lambda_a}{2}\right)} \quad (5)$$

where d_{ab} represents the actual length of the road section calculated using the Haversine distance model; R_{earth} is Earth's average radius, approximately 6371 km; (φ_a, λ_a) and (φ_b, λ_b) are the latitude and longitude of nodes V_a and V_b , respectively.

Considering the randomness of traffic conditions, a time-varying travel time function is defined, as shown in Equations (6) and (7):

$$v_{ab}(\tau) = \bar{v} \cdot \tau_{ab} \quad (6)$$

$$t_{ab}(\tau) = \frac{d_{ab}}{v_{ab}(\tau)} \quad (7)$$

where $v_{ab}(\tau)$ represents the travel speed on the road section between two nodes, influenced by the road resistance coefficient; τ_{ab} denotes the road resistance coefficient for the current section; $\bar{v}$ represents the baseline speed, i.e., the travel speed under ideal conditions; and $t_{ab}(\tau)$ represents the travel time for the current section. It is noteworthy that to reflect the stochastic nature of traffic conditions and reduce the impact of outliers on model outputs, τ_{ab} is defined in this paper as a random number uniformly distributed in the interval [0.1, 1]. Furthermore, to reflect the bidirectional connectivity between nodes, $t_{ab}(\tau)$ is reassigned the value of τ_{ab} during computation.

3.3. Objective Function and Model Constraints

Based on the model assumptions described earlier, the objective function of this study aims to, within scenarios of randomly varying road resistance, minimize the number of station configurations based on the expected target coverage rate, thereby reducing

investment and operational costs while maximizing the coverage effectiveness of the emergency rescue network. Its mathematical formulation is given by Equations (8)–(12):

$$\min f_1(X) = \sum_{l=1}^n X_l \quad (8)$$

$$\max f_2(X) = E[CT(X)] = \frac{1}{S} \sum_{s=1}^S \left[\frac{1}{n} \sum_{m=1}^n I_{ms}(X) \right] \quad (9)$$

$$X_l = \begin{cases} 1, & \text{Set up a repair station at node } V_l \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

$$I_{ms}(X) = \begin{cases} 1, & \min T_{lms} \leq T_{max} \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

$$\eta_{lower} \leq CT(X) \leq \eta_{upper} \quad (12)$$

where X_l is the decision variable for selecting stations from V_{nodes} ; $E[CT(X)]$ is the expected target coverage rate; S represents the set of road resistance scenarios; $I_{ms}(X)$ is the coverage indicator function, determining whether node m is covered in scenario s ; η_{lower} and η_{upper} define the coverage rate boundaries; T_{lms} represents the travel time from node m to station l in scenario s ; T_{max} denotes the maximum arrival time threshold.

To clarify the optimization logic of the proposed CF-SRLP model, it is essential to note that while two objective functions are mathematically presented (Equation (8) for minimizing the number of stations and Equation (9) for maximizing the expected coverage rate), the solution approach is fundamentally a hierarchical or constraint-based optimization rather than a simultaneous multi-objective optimization. The core of the model is to identify the most cost-effective station layout that reliably meets a predetermined service standard under uncertainty. Specifically, the expected coverage reliability, calculated via Monte Carlo simulation across stochastic road resistance scenarios (Equation (9)), serves as the primary performance constraint (as defined by the interval η_{lower} , η_{upper} in Equation (12)). The optimization goal, implemented by the Improved Iterative Greedy Search algorithm, is to minimize the number of stations (Equation (8)) subject to achieving the target coverage reliability (Equation (12)). The algorithm iteratively adds the station yielding the highest marginal coverage gain until the coverage rate meets or exceeds the upper threshold η_{upper} , thereby naturally obtaining the minimal set of stations required for reliable service. This formulation prioritizes finding a robust and economical solution that guarantees a high probability of coverage under stochastic conditions, rather than generating a Pareto frontier between two competing objectives.

Furthermore, based on the assumptions outlined in Section 2.2, the aforementioned mathematical model defines a theoretical upper limit on the performance of emergency response systems. By approximating demand points as network nodes and excluding scheduling conflicts, resource queues, and capacity constraints, the model specifically examines and optimizes the spatial coverage efficiency of station layout under stochastic travel time conditions. This focused scope of study provides clear insights into the fundamental trade-offs between the number of stations, their locations, and coverage reliability. It is worth noting that, since this model does not account for the dynamic factors mentioned above, real-world performance may be more challenging and complex. Therefore, as a foundational planning tool, the model's output can establish a performance baseline for evaluating the impact of additional operational constraints in future comprehensive studies.

3.4. Improved Iterative Greedy Search Algorithm

Iterative Greedy Search [40–42] demonstrates a high approximation ratio when addressing this problem. Furthermore, its parallel computing capability endows it with high computational efficiency in solving multi-scenario problems. Building on this and considering the high-dimensional combinatorial optimization characteristics of the model, this paper designs an Improved Iterative Greedy Search algorithm. Its core idea is to generate multiple configuration schemes that satisfy the constraints by iteratively adding the station providing the greatest marginal benefit. The algorithmic process is illustrated in Figure 5.

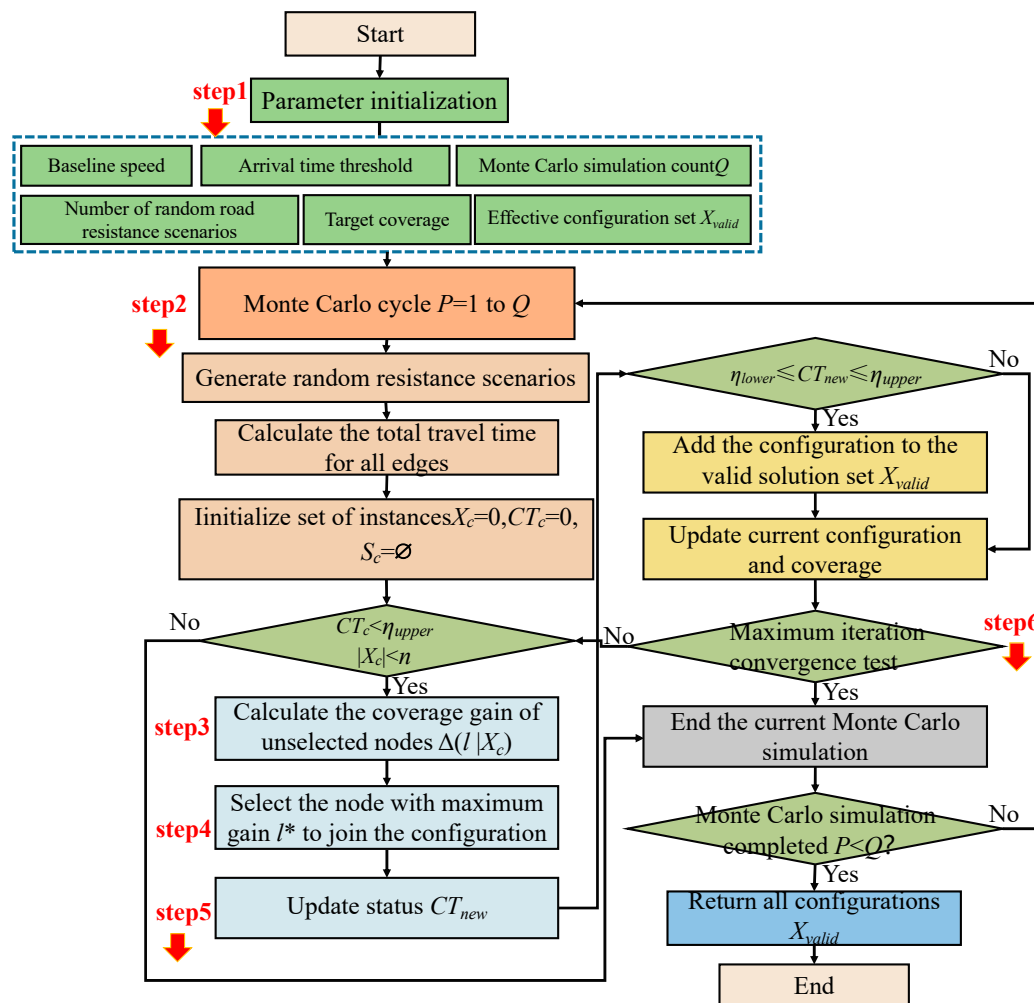


Figure 5. Schematic diagram of the algorithm flow.

The algorithm flow can be divided into 6 steps, detailed as follows.

1. Parameter initialization and input settings. Define the baseline speed, arrival time threshold, number of random road resistance scenarios, number of Monte Carlo simulations Q , target coverage rate interval, and initialize the set of valid configurations X_{valid} .
2. Monte Carlo simulation loop framework. The outer loop employs the Monte Carlo simulation method to generate statistically independent random road resistance scenario sets. For each simulation instance, a random road resistance coefficient vector is first generated. Subsequently, the current configuration state is initialized: the set of selected stations X_c and the set of covered nodes S_c are empty, and the current coverage rate CT_c is set. This initialization provides a clear starting state for the subsequent greedy optimization.

3. In each iteration, the algorithm evaluates the marginal contribution of all unselected nodes. For each node $V_i \notin S_c$ under evaluation, its marginal coverage gain $\Delta(l|X_c)$ is calculated. This gain function is defined as the expected number of currently uncovered nodes that node V_i can cover across all random scenarios. The mathematical expression is given by Equation (13):

$$\Delta(l|X_c) = \frac{1}{S} \sum_{s=1}^S \sum_{m=1}^n I\{T_{lms} \leq T_{max} \wedge I_{ms}(X_c) = 0\} \quad (13)$$

4. Optimal node selection strategy. Based on the calculated marginal gains, a greedy strategy selects the current optimal node. The selection criterion is given by Equation (14):

$$l^* = \arg \max_{l \notin S_current} \Delta(l|X_current) \quad (14)$$

5. Coverage rate calculation and feasibility verification. After updating the configuration, the coverage rate indicator CT_{new} needs to be recalculated. Subsequently, feasibility verification is performed to check if the constraint $\eta_{lower} \leq CT_{new} \leq \eta_{upper}$ is satisfied. If this condition holds, the current configuration is a feasible solution and is added to the valid configuration set X_{valid} .
6. Termination condition and output of valid configuration set. First, check if the upper coverage limit is reached $CT_c \geq \eta_{upper}$, to terminate the iterative process for the current Monte Carlo simulation. Second, determine if the Monte Carlo simulation loop is complete, and then output the valid configuration set X_{valid} .

Thus, the model construction section of this paper is now complete.

4. Multi-Dimensional Performance Evaluation Framework

To scientifically assess and compare the comprehensive effectiveness of different location strategies and provide a quantitative basis for resource allocation decisions, this study adopts a multi-dimensional evaluation system to conduct a comprehensive performance assessment of gas emergency repair station location strategies [32,35,43]. Specifically, within the Monte Carlo simulation framework, the performance of each location strategy R is tested under S stochastic road resistance scenarios. The following evaluation metrics are defined, as shown in Equations (15)–(19):

1. Coverage compliance rate

$$A_c(R) = \frac{1}{S} \sum_{s=1}^S I\{C_{R,s} \geq C_{target}\} \quad (15)$$

where $C_{R,s}$ denotes the node coverage rate of strategy R under scenario s , C_{target} is the target coverage rate threshold. This metric reflects the probabilistic assurance level of service reliability for the location scheme.

2. Average arrival time compliance rate

$$A_t^{avg}(R) = \frac{1}{S} \sum_{s=1}^S I\{T_{avg,R,s} \leq T_{target}\} \quad (16)$$

where $T_{avg,R,s}$ is the average arrival time across all nodes for strategy R under scenario s , T_{target} is the emergency response time standard. This metric characterizes the timeliness of the scheme under typical operating conditions.

3. Maximum arrival time compliance rate

$$A_t^{max}(R) = \frac{1}{S} \sum_{s=1}^S I\{T_{max,R,s} \leq T_{target}\} \quad (17)$$

where $T_{max,R,s}$ is the arrival time for the farthest node under strategy R and scenario s . This metric reflects the service equity and robustness of the scheme under extreme operating conditions.

4. Composite score

$$WCS(R) = \omega_c \cdot A_c(R) + \omega_{avg} \cdot A_t^{avg}(R) + \omega_{max} \cdot A_t^{max}(R) \quad (18)$$

The weight configuration in this paper is set as $\omega_c = 0.4$, $\omega_{avg} = 0.3$, $\omega_{max} = 0.3$, reflecting a decision-making preference that prioritizes coverage reliability while also valuing response efficiency.

5. Station efficiency

$$SE(R) = \frac{WCS(R)}{N_R} \quad (19)$$

where N_R is the number of stations corresponding to location strategy R . This metric quantifies the comprehensive effectiveness per unit station, providing a basis for cost-benefit analysis.

5. Case Study and Empirical Analysis

5.1. Study Area and Topological Network Construction

City H is a pivotal central city within the Yangtze River Delta urban agglomeration and comprises six main administrative districts: Shangcheng, Xiacheng, Gongshu, Jianggan, Binjiang, and Xihu. The city combines rich historical and cultural heritage with highly modernized urban functions, as illustrated in Figure 6. The spatial structures of its main urban road network and gas pipeline network exhibit strong coupling and interdependency.

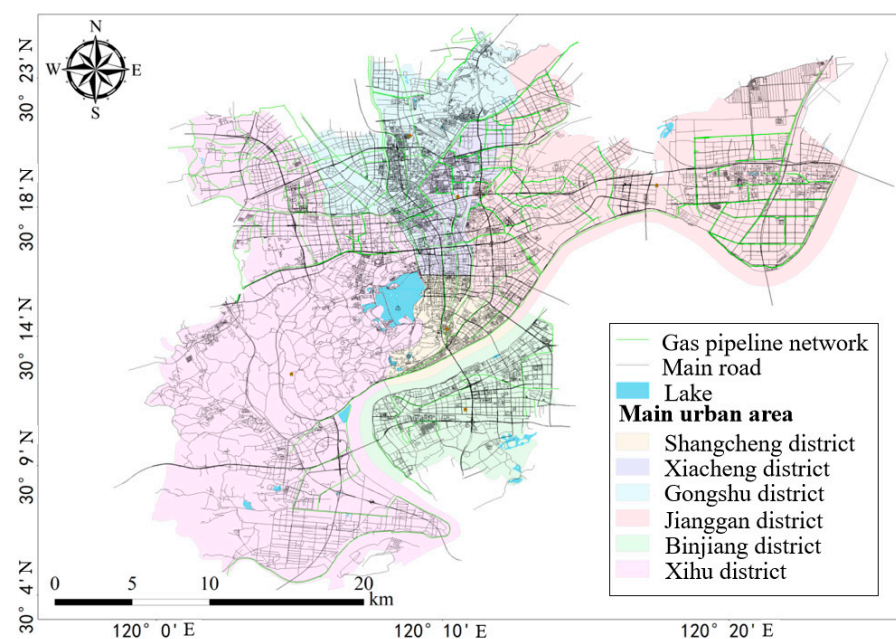


Figure 6. Complex road network data of City H.

The road network displays heterogeneous characteristics, with high connectivity in the core urban area and relatively weaker connections in peripheral zones. This structural feature makes certain regions prone to recurrent congestion over multiple periods. The gas pipeline network generally follows the alignment of major arterial roads, and pipeline density is positively correlated with road density in spatial distribution. Furthermore, the pipeline network presents a clear center–periphery gradient, with dense coverage and high connectivity in the core built-up area and relatively sparse distribution in newly developed or peripheral areas. This pattern reflects the city’s development stages and the spatiotemporal sequence of infrastructure investment.

Given these characteristics, the main urban area of City H was selected as a representative case to empirically validate the proposed candidate-free location optimization method for gas emergency repair stations. Road data downloaded from OpenStreetMap was first imported into ArcGIS10.8. Considering the layout of the gas pipeline network, 143 main urban arterial roads with gas pipelines laid along them were selected for analysis. The selected road data were subsequently imported into MATLAB2023b, where identification, simplification, and deduplication procedures were performed according to Equations (1)–(4). This process resulted in a fully connected topological network consisting of 498 nodes and 1192 links, as shown in Figure 7.

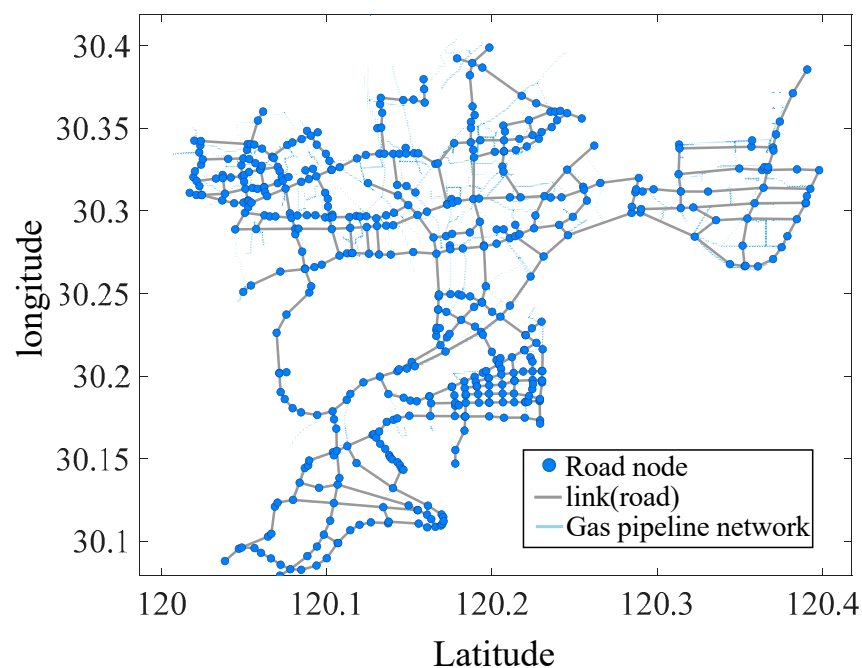


Figure 7. Demand point-road network topology of City H.

The simplified processing aims to clearly represent connectivity relationships between nodes in order to facilitate model analysis and computation. In practical applications, the network can be further refined by incorporating additional geographical attributes based on specific operational requirements.

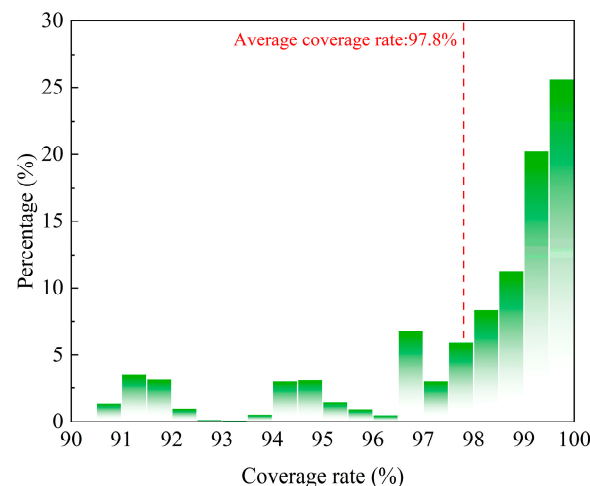
Based on the constructed topological network of City H, the proposed location model was applied for numerical simulation. A total of 20,000 stochastic road resistance scenarios were generated, and 2125 station configuration schemes were evaluated. The core simulation parameter settings are presented in Table 2. To enable systematic comparison of performance, the empirical analysis is conducted from three perspectives: coverage rate distribution, station selection characteristics, and arrival-time performance of demand points.

Table 2. Parameter settings.

Parameter Name	Baseline Speed (km/h)	Target Time (h)	Number of Road Resistance Scenarios	Number of Monte Carlo Simulations	Target Coverage Rate (%)
Parameter Symbol Value	$\bar{v}$ 60	T_{max} 0.5	numScenarios 100	numMonteCarlo 200	$CT(X)$ [90, 100]

5.2. Multi-Dimensional Analysis of Coverage Rate Distribution

The distributional characteristics of system coverage performance under stochastic road resistance are first examined to evaluate robustness. As shown in Figure 8, the coverage rate distribution exhibits a pronounced right-skewed pattern, with probability density concentrated in the high-value interval of 96% to 100%. This indicates that the configurations generated by the proposed method achieve high target coverage under the vast majority of stochastic traffic conditions. The extremely low probability density in lower coverage intervals suggests a minimal risk of coverage failure in practical implementation. These results confirm the robustness of the proposed method under stochastic road resistance.

**Figure 8.** Probability density distribution of the coverage rate.

The central tendency and dispersion of coverage rates are further analyzed to assess stability across scenarios. As illustrated in Figure 9, the mean coverage rate reaches 97.77%, demonstrating strong central tendency across all simulated scenarios. Indicators of dispersion remain low, with a standard deviation of 2.48%, a range of 9.59%, and an interquartile range of 2.54%. The coefficient of variation is 2.54%, indicating limited relative variability in coverage performance. These findings show that the optimized solutions not only achieve high effectiveness but also maintain stable and reproducible performance despite stochastic traffic disturbances.

Figure 10 presents the convergence behavior of the solution process. Both the average number of stations and the average coverage rate increase rapidly in the early phase of the configuration sequence and subsequently stabilize. The average station number converges to approximately 9.8, and the average coverage rate stabilizes around 97.8%. This suggests that the optimization process is relatively insensitive to random initial conditions or search paths and consistently yields reproducible near-optimal solutions. For a given input configuration, the model provides a clear station scale with predictable performance expectations, offering reliable quantitative support for decision-making.

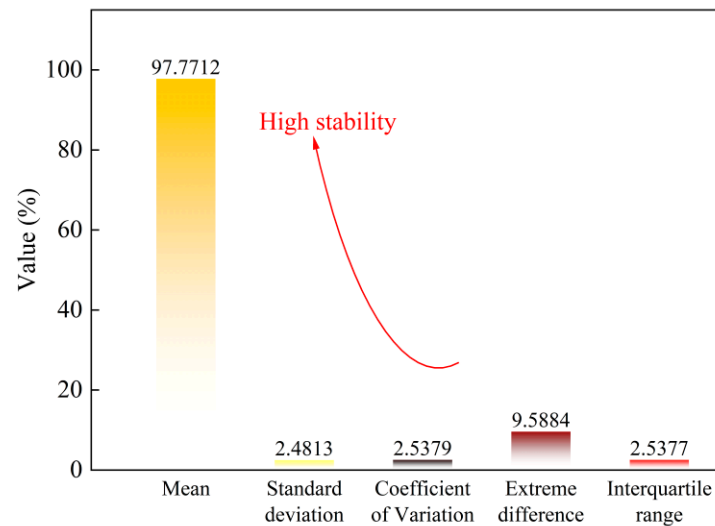


Figure 9. Statistical characteristics and stability analysis of the scheme's coverage rate.

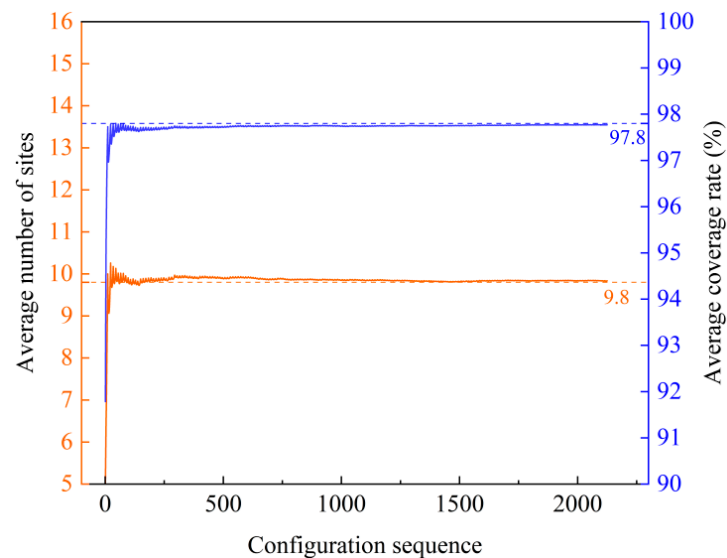


Figure 10. Convergence of average station number and coverage rate.

The relationship between station scale and coverage performance is subsequently investigated to understand marginal investment effects. Figure 11 reveals the quantitative relationship between the number of stations and system coverage rate. As the number of stations increases from 5 to 17, the mean coverage rate rises from 91.5445% to 99.7952%, and the dispersion of coverage rates gradually decreases. This confirms a strong positive association between station scale and coverage performance. However, the increase is not linear. When the station count rises from 5 to 11, the mean coverage rate increases substantially from 91.5445% to 99.2234%. Beyond 11 stations, further expansion to 17 stations produces only a 0.5718% increase, indicating a plateau phase. This demonstrates a clear pattern of diminishing marginal returns and suggests the existence of a cost-effective optimal range for station scale under resource constraints. When aiming for coverage rates approaching 100%, careful consideration of marginal investment benefits is therefore required.

Overall, the proposed method generates high-coverage configurations with stable performance in uncertain environments. Although increasing the number of stations improves coverage, the growth follows a diminishing marginal return pattern, providing important insights for balancing performance improvement and cost efficiency.

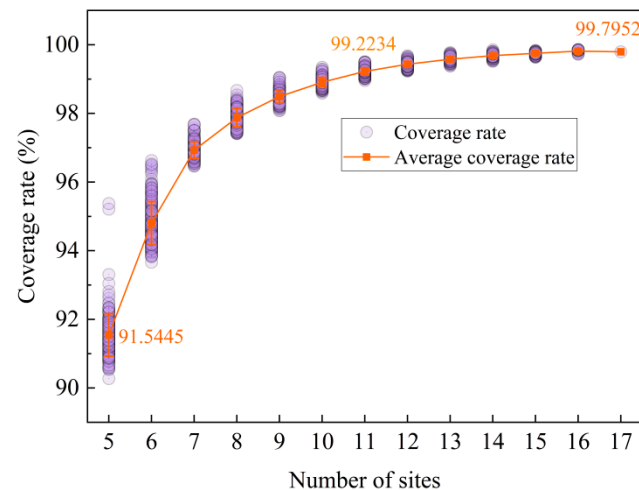


Figure 11. Relationship between the number of stations and the coverage rate.

5.3. Overall Analysis of Station Selection

The spatial stability and robustness of station selection under stochastic conditions are next examined. Figure 12 presents a heatmap of station selection frequency under stochastic road resistance scenarios. The results reveal pronounced spatial heterogeneity in selection frequency, forming a hierarchical pattern consisting of a limited number of high-frequency core nodes and a large number of low-frequency nodes. High-frequency stations are concentrated in specific geographic areas that coincide with major road network hubs or potential high-demand regions, reflecting the spatial consistency and internal logic of the model's location decisions. The frequency distribution represents a probabilistic expression of decision robustness under uncertainty and provides data-driven support for identifying and prioritizing core stations.

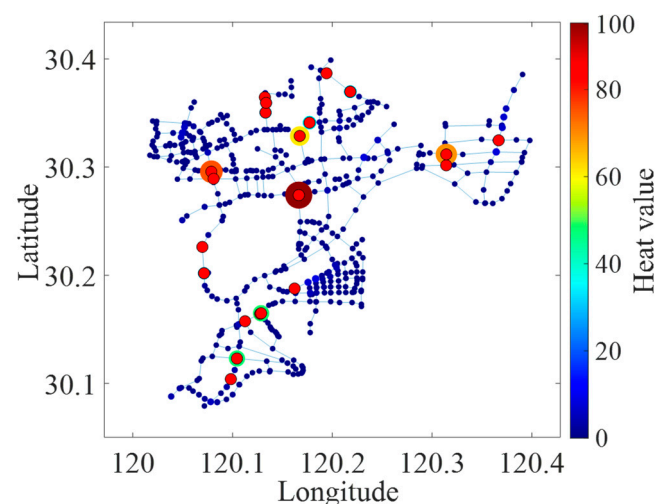


Figure 12. Station selection frequency heatmap.

Figure 13 further illustrates the ranking of station selection probabilities and reveals a clear long-tailed distribution. A small number of nodes, including 260, 91, and 463, exhibit significantly higher selection probabilities and form the “head” of the distribution, while most nodes lie in the lower-probability “tail.” This pattern indicates that a limited number of structurally important hubs consistently play a central role across scenarios, whereas most stations are scenario-dependent and become important only under specific road resistance conditions.

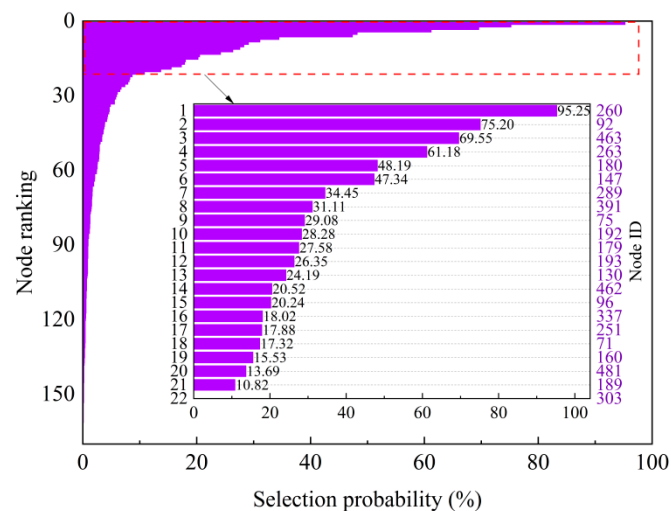


Figure 13. Ranking of station selection probabilities.

These results demonstrate that the model exhibits consistent spatial preferences across repeated stochastic simulations. This consistency verifies the robustness of the optimization results and provides practical guidance for phased implementation. High-probability core stations can be prioritized to establish a stable service backbone, while medium- and low-probability stations may be incorporated progressively according to budget constraints and resilience requirements.

5.4. Comparative Analysis of Arrival Conditions

System response time reflects not only overall service efficiency but also vulnerability under extreme conditions. A comparison between average arrival time and maximum arrival time therefore provides insight into both routine performance and tail risk. As illustrated in Figure 14, average arrival times are relatively concentrated, primarily within the interval of 6 to 20 min, indicating that under most stochastic scenarios, the system maintains a satisfactory response level. In contrast, maximum arrival times display a much broader dispersion, mainly concentrated between 30 and 45 min. Only 12.0414% of maximum arrival times fall within the rescue time threshold. This contrast reveals that although the system performs well on average, substantial delays persist in extreme scenarios. This further indicates that the risk of extreme delays is an inherent characteristic of spatially random layouts. In practice, this risk is also influenced by operational factors such as scheduling conflicts and resource queues. Therefore, improving average efficiency alone is insufficient; controlling maximum response time becomes equally important.

The effect of expanding station scale further clarifies this distinction. As shown in Figure 15, increasing the number of stations consistently reduces both average and maximum arrival times. The average arrival time decreases from 16.8845 min with 5 stations to 8.7288 min with 17 stations, demonstrating steady gains in overall service efficiency. Meanwhile, the maximum arrival time declines from 48.7833 min to 31.7817 min, confirming that resource expansion partially mitigates extreme delays. Nevertheless, even at 17 stations, the average level of maximum arrival time remains above the rescue time threshold. This suggests that simply increasing station quantity cannot eliminate timeout risks under unfavorable traffic scenarios.

The dynamics of improvement rates further reveal how performance gains evolve with scale. As shown in Figure 16, the improvement rate of average arrival time reaches its peak of 13.3052% when the station number is 7. After this point, although efficiency continues to improve, the marginal benefit gradually declines. In contrast, the improvement rate of maximum arrival time increases more slowly and peaks later at 6.6413% when the station

number reaches 11. This lag indicates that improving worst-case performance requires a larger base of resources and that its benefits are released more gradually. Notably, when the number of stations increases to 17, the improvement rate of maximum arrival time becomes negative (-2.1582%). This indicates that excessive expansion may even reduce efficiency in extreme scenarios, potentially due to increasing network complexity.

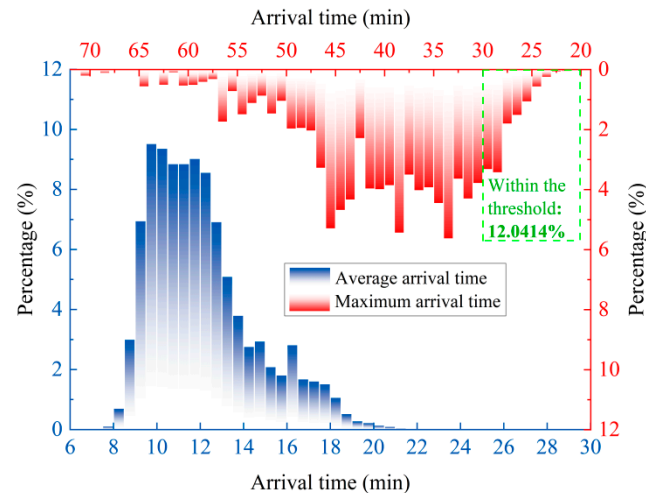


Figure 14. Comparison of arrival time distributions.

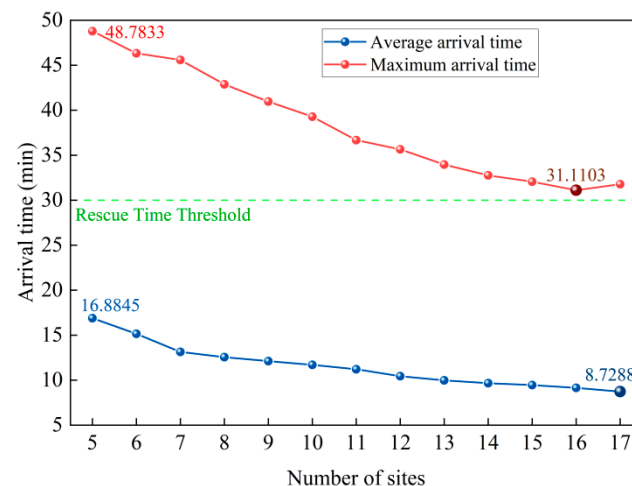


Figure 15. Variation in arrival times with the number of stations.

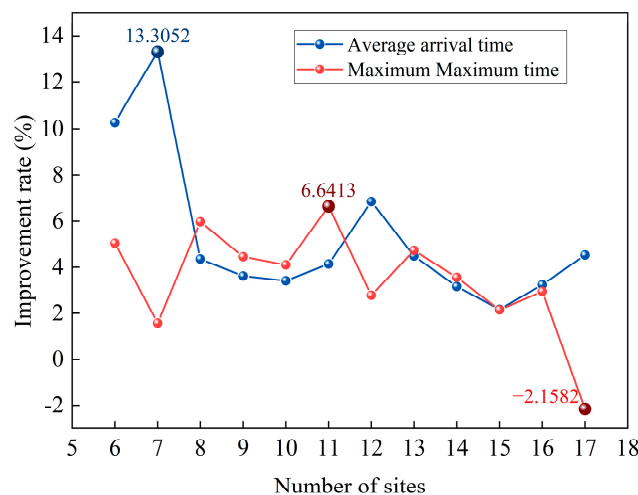


Figure 16. Variation in arrival time improvement rates with the number of stations.

A broader view of response time distributions reinforces these observations. Figure 17 shows that the medians of both average and maximum arrival times decrease steadily as the number of stations increases, confirming the positive impact of resource investment on response speed. However, the rate of reduction becomes progressively flatter once the number of stations exceeds 11, visually reflecting diminishing marginal returns. At the same time, the interquartile ranges narrow, indicating enhanced stability and reduced variability in response performance. Despite this improvement in central tendency and dispersion, numerous high-value outliers for maximum arrival time remain even at 17 stations. This demonstrates that while scale expansion improves average performance and stability, it does not fully remove extreme delay risks.

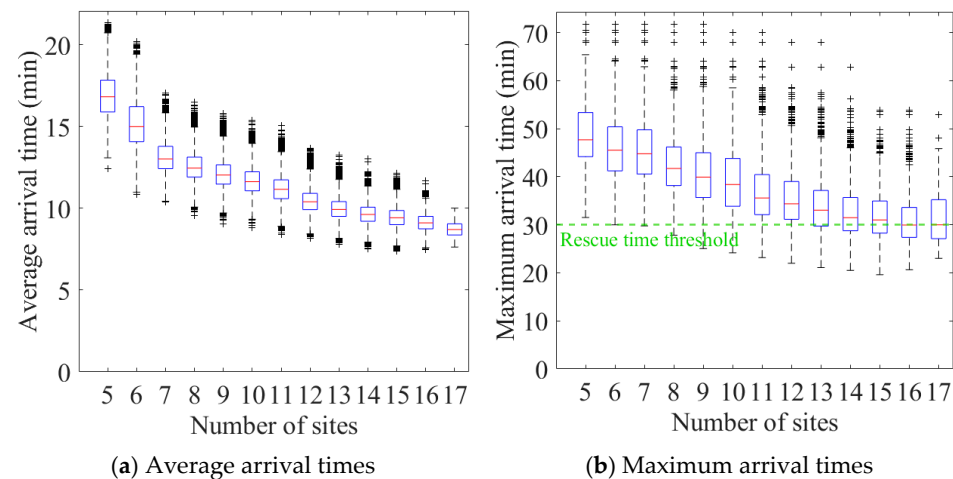


Figure 17. Box plots comparing arrival times.

These findings indicate that while increasing the number of stations enhances average performance and stability, it cannot fundamentally resolve extreme delay risks. Effective planning should therefore combine scale optimization with structural and managerial improvements, determining a cost-effective base scale and complementing it with refined dispatching and bottleneck management strategies.

5.5. CF-SRLP Performance Analysis

In order to verify the superiority of the candidate point free random road resistance location model (CF-SRLP) proposed in this paper, this section compares it with two classical benchmark location methods: (1) Deterministic location method, that is, the road resistance coefficient is fixed to the average value ($\tau = 0.6$) in the model, ignoring its randomness to optimize; (2) Euclidean distance location method, a classical heuristic method for coverage judgment and location based on linear distance between nodes. Under the same network and demand point settings, we use the same improved iterative greedy algorithm to generate the optimal layout of 10 stations for the three methods, and evaluate its performance in 200 independent random road resistance scenarios. The comparison results are shown in Table 3.

Table 3. Performance comparison of different location methods.

Model	Average Coverage (%)	Average Arrival Time (min)	Average Maximum Arrival Time (min)
CF-SRLP	97.5371	14.1739	38.7527
Deterministic location	71.9378	23.5073	58.2002
Euclidean distance location	22.4719	54.0384	108.2113

Specifically, the CF-SRLP method proposed in this paper is significantly better than the deterministic location method and the Euclidean distance location method in the three core performance indicators of average coverage, average arrival time and average maximum arrival time. This significant performance improvement is directly attributed to the core innovation of the CF-SRLP model: the explicit integrated Monte Carlo random scene simulation is used to optimize the generated site layout, so that the generated site layout can actively avoid the vulnerable positions that are easy to fail under random traffic congestion, thus systematically enhancing the robustness of the emergency service network. The deterministic method ignores the traffic uncertainty, and the Euclidean distance method is separated from the actual road network topology, and its optimal layout is fragile under random pressure test.

To evaluate the emergency response effectiveness of the optimized site selection schemes, this section compares the optimized site schemes obtained from the developed site selection optimization method with the existing site layout in City H.

The site schemes are presented in Table 4. Scheme 1 selects the top 6 sites based on their probability rankings, maintaining the same number of sites as the existing layout in City H (Scheme 2). Based on the two defined site layout schemes, simulation analyses were conducted across 10 sets of random traffic obstruction scenarios. In each traffic obstruction scenario, 50 accident nodes were randomly generated, resulting in a total of 500 simulated accidents. The spatial distribution of the two schemes is shown in Figure 18.

Table 4. Site Configuration Settings.

Scheme	Description	Number of Sites	Site Number
1	A set of sites selected based on the method described in this paper	6	[260; 92; 463; 263; 180; 147]
2	Current site location combinations in City H	6	[87; 284; 224; 276; 346; 457]

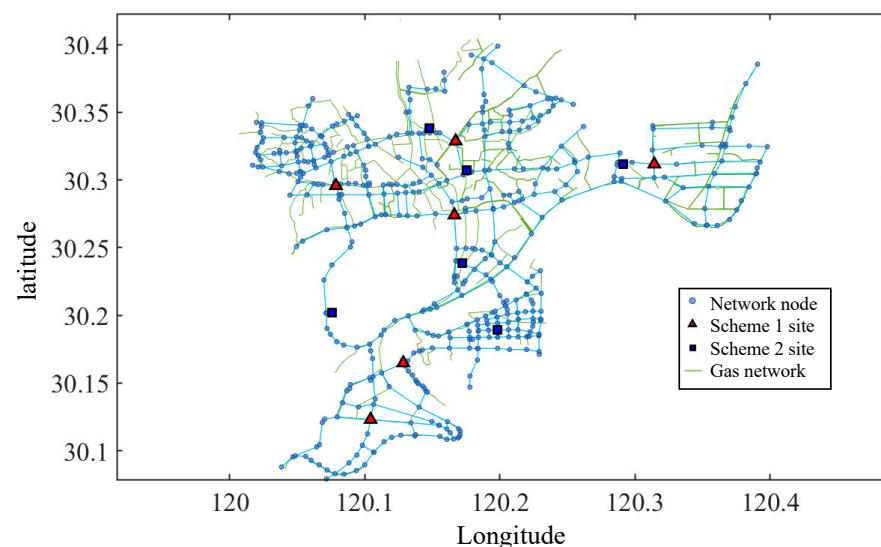


Figure 18. Spatial distribution diagram of two schemes.

As shown in Figure 19, a comparative analysis of the response time distributions for Scheme 1 and Scheme 2 reveals significant differences between the two scenarios in terms of the degree of time concentration, the shape of the distribution, and the presence of a long-tail characteristic. Within the critical response window of [10,20] minutes, the percentage distribution for Scheme 1 is significantly higher than that of Scheme 2. This

indicates that the optimized layout enables response times for a greater number of incidents to be controlled within this optimal timeframe, demonstrating superior timeliness under routine conditions. However, for time intervals of 20 min or longer, Scheme 2 accounts for a slightly higher proportion than Scheme 1, suggesting that under the current layout, the risk of prolonged delays is relatively higher. From the overall distribution pattern, the curve for Scheme 1 exhibits a steeper and more concentrated profile, indicating a tighter distribution of response times and greater stability; the curve for Scheme 2 is relatively flatter and more dispersed, reflecting significant fluctuations in its response performance. In summary, the optimized solution not only demonstrates superior performance in reducing average response times but also enhances the consistency and reliability of emergency responses, thereby validating the effectiveness of the model optimization from the perspective of probability distribution.

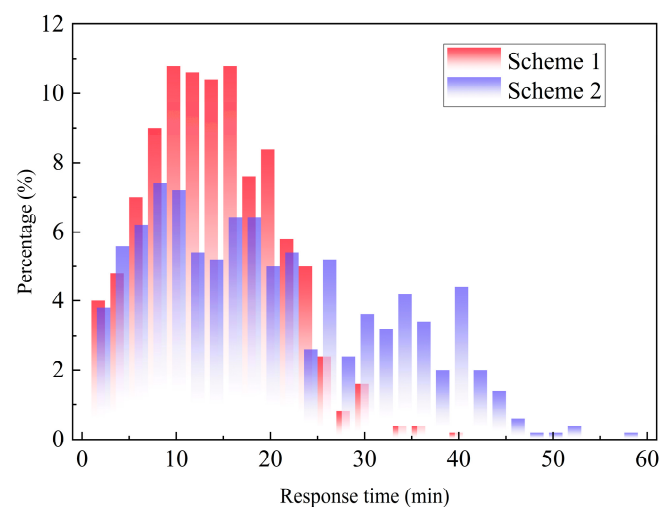


Figure 19. Overall response time distribution across different solutions.

As shown in Figure 20, the response time distributions of Scheme 1 and Scheme 2 exhibit systematic differences across the 10 random traffic congestion scenarios. The overall distribution of response times for Scheme 1 across all scenarios is relatively concentrated, with the main density interval falling within the range of [10,30] minutes, and the vast majority of data points lying below the 30 min response threshold. This indicates that it is capable of maintaining a relatively stable and efficient emergency response level under various uncertain traffic conditions. In contrast, the response time distribution of Scheme 2 is significantly more dispersed, with a wider data range; in many scenarios, a considerable proportion of response times exceed the 30 min threshold, and both the median and upper quartile are generally higher than those of Scheme 1. This indicates that its response performance is susceptible to fluctuations in road obstructions, posing a higher risk of delays, and further highlights the clear advantage of Scheme 1 in controlling response time overruns. Therefore, as confirmed by this distribution from both multi-scenario and probability density perspectives, the model-optimized layout can systematically compress the response time distribution, thereby reducing the risk of delays and ultimately effectively enhancing the overall response performance and robustness of the emergency system.

As shown in Figure 21, the system reliability of Scheme 1 consistently exceeds that of Scheme 2 across a wide range of time threshold scenarios. Under the critical 30 min emergency response threshold scenario, the reliability of Scheme 1 reaches 99.00%, while that of Scheme 2 is only 77.80%, a difference of 21.20%. This demonstrates that the layout schemes generated by the site selection optimization method proposed in this study can improve the system's assurance rate of completing tasks within the standard response time.

From the dynamic changes in the curves, the reliability of Scheme 1 rises rapidly as the threshold increases and reaches a stable state earlier, indicating that its emergency response capacity is released more concentratedly and efficiently; in contrast, the reliability of Scheme 2 grows more gradually and has not fully converged even by the 50 min mark, revealing a certain degree of lag in its response process and, relatively speaking, lower efficiency in the utilization of time resources. These results further confirm that optimization methods accounting for road obstruction uncertainty not only systematically reduce the average response time but also significantly enhance the system's certainty of successful response within a specified time window, thereby providing superior timeliness assurance and operational reliability in actual emergency management.

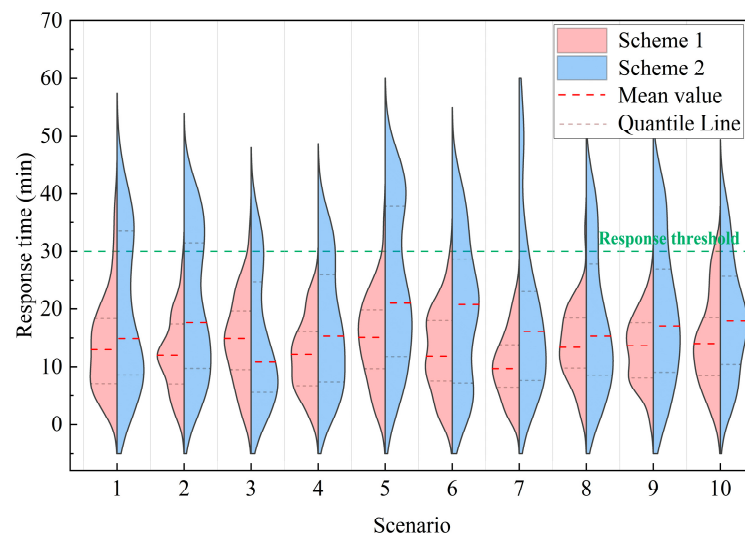


Figure 20. Response time distribution of two solutions across different scenarios.

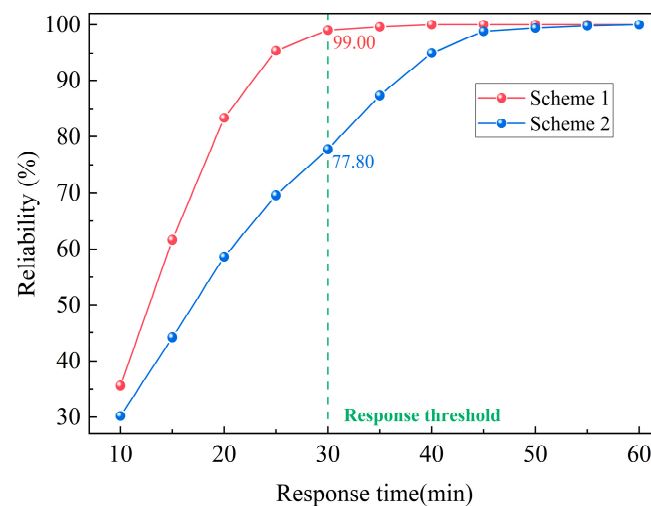


Figure 21. Reliability comparison at different time thresholds.

5.6. Feasibility Analysis of Station Selection Strategies

To further highlight the decision-making advantages of the constructed location model under the constraints of stochastic road resistance and the absence of fixed candidate points, this study systematically designs five progressively expanded location strategies, *R1* to *R5*, for comparative validation based on the station selection probability ranking output by the model, as shown in Table 5. These strategies are evaluated under the same 1000 stochastic road resistance scenarios. This controlled approach allows for a fair comparison of the

performance of strategies at different scales under uncertainty, revealing the rationality and robustness of the station sequence recommended by the proposed method.

Table 5. Station selection strategy settings.

Strategy R	Number of Stations	Station IDs	Number of Stochastic Road Resistance Scenarios
R1	5	[260; 92; 463; 263; 180]	1000
R2	8	[260; 92; 463; 263; 180; 147; 289; 391]	
R3	10	[260; 92; 463; 263; 180; 147; 289; 391; 75; 192]	
R4	12	[260; 92; 463; 263; 180; 147; 289; 391; 75; 192; 179; 193]	
R5	17	[260; 92; 463; 263; 180; 147; 289; 391; 75; 192; 179; 193; 130; 462; 96; 337; 251]	

As shown in Figure 22, the five location strategies exhibit distinct distribution characteristics in terms of coverage rate, average arrival time, and maximum arrival time.

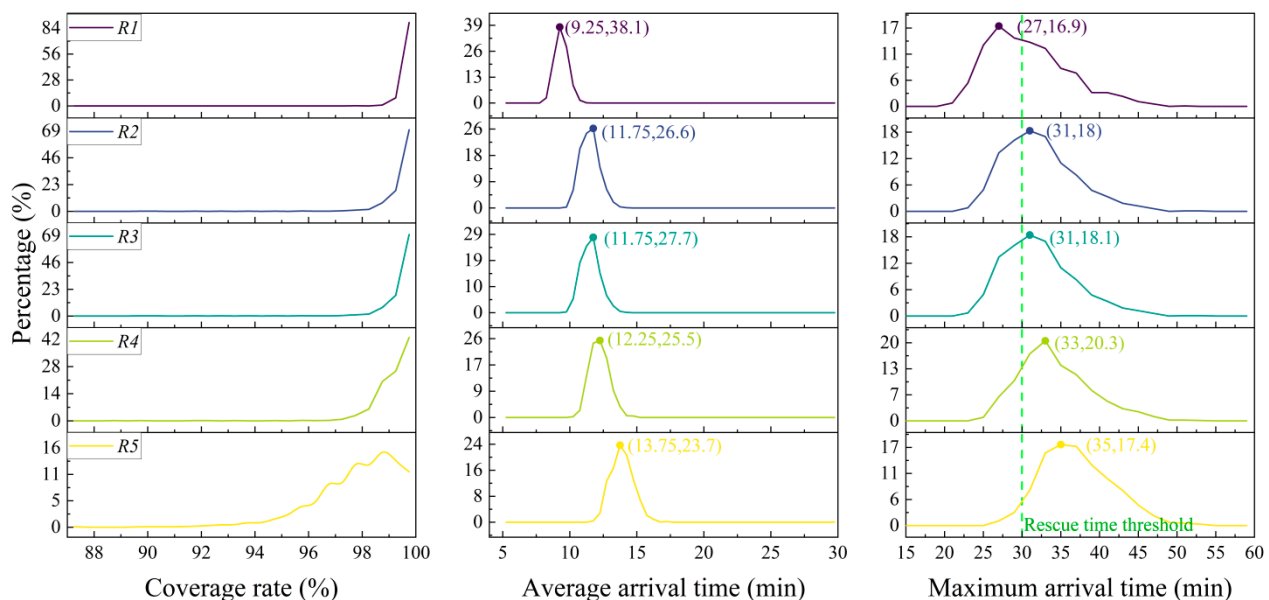


Figure 22. Distribution of coverage rate, average arrival time, and maximum arrival time under different strategies.

Regarding coverage rate, all five strategies achieve coverage above 90%, but their performance boundaries and improvement paths differ significantly. The coverage for R5 is concentrated in the [99%, 100%] range, with nearly 84% of simulated scenarios reaching a high coverage rate above 99%, indicating extremely high stability and excellent performance of the system. In contrast, for R2 to R4, the proportion of coverage in the [99%, 100%] range decreases noticeably, while it steadily increases in the [98%, 99%] range, suggesting that system response performance becomes more susceptible to road resistance. R1 shows coverage distributed in [95%, 99%], where the impact of stochastic road resistance on system performance is even more pronounced. This visually demonstrates that under stochastic road resistance, resource scale is the fundamental guarantee for achieving high and stable coverage.

In terms of average arrival time, all strategies meet the rescue time threshold. From R1 to R5, the distribution shifts leftward from 12–17 min to 9–12 min, and peak values decrease from 13.15 min to 9.25 min, confirming improved response efficiency with expansion. The small difference between R3 and R4 further reflects diminishing marginal returns.

Regarding maximum arrival time, the curves for all strategies show significant right-skewness, indicating the presence of high-risk scenarios with substantial delays in the interval beyond the arrival time threshold. Observation reveals that the peak of the maximum arrival time decreases from 35 min for R1 to 27 min for R5. However, the overall interval remains stable within [25 min, 60 min] without a clear tendency to concentrate towards the peak. This suggests that increasing resource scale can indeed improve the response capability of stations under extreme road resistance scenarios to some extent, but significant tail risks persist.

To scientifically evaluate and compare the comprehensive effectiveness of different location strategies, performance tests were conducted on R1–R5 based on the previously described multi-dimensional performance evaluation method, as shown in Figures 23 and 24.

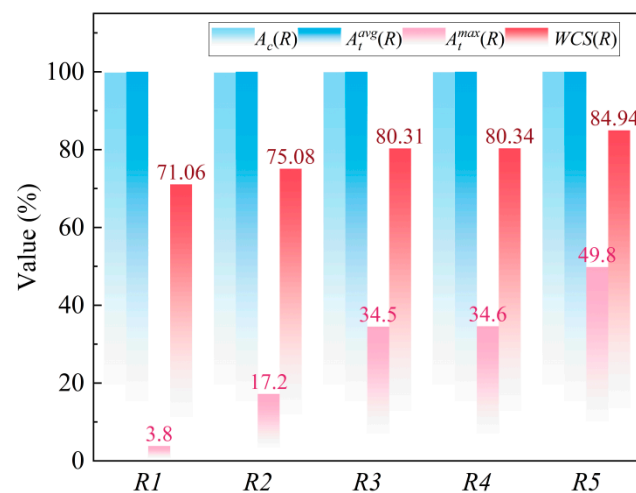


Figure 23. Comprehensive evaluation of the five strategies.

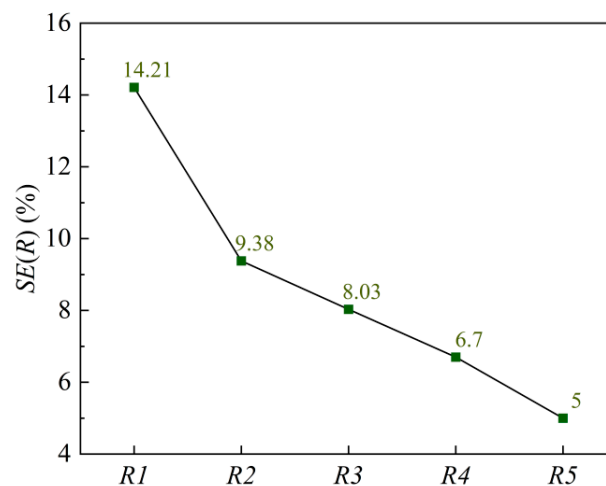


Figure 24. Station efficiency evaluation of the five strategies.

Figure 23 shows that the $A_c(R)$ and $A_t^{avg}(R)$ for R1 to R5 are all close to 100%, indicating their ability to provide good basic service in most regular scenarios. However, $A_t^{max}(R)$ increases significantly from 3.8% to 49.8%, becoming the most critical factor constraining the comprehensive response effectiveness of the gas pipeline network system. This further confirms that the station selection schemes output by the proposed method can meet the general response requirements of City H for gas incidents and have a certain promoting effect on response effectiveness in extreme scenarios. Therefore, decision-makers can use this method to first ensure the targets of “coverage rate” and “average arrival time”, and

subsequently seek strategies such as dynamic dispatch, integrated maintenance, and route optimization to tackle the challenge of service reliability in extreme scenarios.

From Figure 24, it can be found that $WCS(R)$ decreases from 14.21% for $R1$ to 5% for $R5$. This indicates that relying solely on increasing the number of stations to improve system performance will lead to the continuous dilution of the average output per unit resource. Therefore, planning decisions should avoid falling into the trap of “scale-only theory” and instead seek the optimal balance between “acceptable comprehensive performance” and “affordable resource efficiency”.

In summary, the station selection schemes output by the proposed method can systematically guarantee basic service effectiveness, keeping both $A_c(R)$ and $A_t^{avg}(R)$ stable at a high level close to 100%. This ensures reliable service in the vast majority of stochastic road resistance scenarios. Crucially, this method effectively optimizes the potential for responding to extreme scenarios by selecting a probabilistically ranked set of sites, significantly improving $A_t^{max}(R)$ compliance rate and markedly enhancing the system’s robustness. At the same time, the marginal benefit curve provides a precise cost-effectiveness boundary for decision-making.

6. Discussion

This study addressed the problem of locating gas emergency repair stations under stochastic road resistance by developing a candidate-free location optimization framework. The case study of City H demonstrates that the proposed model achieves an average coverage rate of 97.77% across 20,000 stochastic scenarios within the specified response time threshold. The results confirm that increasing the number of stations improves both coverage and arrival-time performance; however, the improvement follows a clear law of diminishing marginal returns.

From a theoretical perspective, the study extends classical covering location models by eliminating reliance on predefined candidate sites and incorporating stochastic traffic conditions. Compared with deterministic approaches [12–15,25–29], which, although effective under stable conditions, may restrict the global search space and underestimate uncertainty effects. To overcome these limitations, this research performs a network-wide optimized search without predefined candidate sites, formulating the problem as a covering optimization problem under stochastic conditions.

To characterize and solve this problem, a candidate-free location optimization model incorporating stochastic road resistance was constructed. The model advances existing approaches in three main respects. First, it eliminates dependence on predefined candidate points and enables a global search across the entire road network domain, extending coverage location frameworks such as MCLP and LSCP [25–29] into a candidate-free context. Second, it explicitly integrates time-varying traffic randomness through stochastic road resistance coefficients, thereby responding to calls in the literature for incorporating congestion and disruption uncertainty into facility location models [7–9,22]. Compared with gradual coverage and cooperative covering extensions [20,21,30,31], the present model shifts the emphasis from deterministic coverage radius assumptions toward probabilistic coverage reliability under multi-scenario simulation. Third, it proposes a hybrid solution framework combining an improved iterative greedy search algorithm with Monte Carlo simulation. This approach aligns with heuristic and metaheuristic developments in large-scale facility location and emergency planning problems [35,36,40–42], while specifically adapting them to multi-scenario stochastic evaluation. The key distinction between this candidate-free paradigm and candidate-based site selection models that rely on empirical rules lies in its optimization logic and the way it interacts with empirical constraints. In the traditional candidate-based paradigm, empirical judgments such as land availability and

construction costs are used to pre-screen a discrete set of candidates, and the optimization process is confined to this subspace. In contrast, the CF-SRLP framework proposed in this paper first performs global optimization in the complete network node space with coverage reliability as the core objective, thereby identifying layout schemes with theoretically superior performance. Subsequently, specific, heterogeneous real-world constraints (such as the actual unavailability of a particular node) can be applied to these high-performance solutions as filtering or adjustment conditions during post-optimization. This sequence of “optimizing performance first, then adapting to constraints” ensures that decision-makers can make trade-offs based on a clear spatial efficiency frontier, rather than allowing preliminary empirical screening to prematurely and potentially suboptimally constrain the solution space.

The case study of City H demonstrates the effectiveness and stability of the proposed framework. Across 20,000 stochastic road resistance scenarios, the optimized station layout achieves an average coverage rate of 97.77% within the specified response time threshold. Coverage performance remains highly stable, indicating that the proposed method is robust to traffic variability. These findings are consistent with resilience-oriented planning studies [3,4,43], which emphasize that infrastructure robustness should be evaluated not only under typical conditions but also across a wide range of uncertain scenarios. The present study contributes to this perspective by quantifying coverage reliability and arrival-time compliance rates under large-scale stochastic simulations.

The results further confirm a positive correlation between the number of stations and both coverage rate and response time performance, echoing conclusions from classical maximal covering and emergency station planning studies [12–15,26]. However, the analysis clearly reveals a law of diminishing marginal returns: after reaching a certain station scale, further expansion produces only limited improvement in coverage and arrival time indicators. This finding complements existing multi-objective and hierarchical siting research [15,36] by providing empirical evidence, under stochastic traffic conditions, that optimal resource allocation requires balancing performance gains against efficiency dilution. In particular, the decline in station efficiency indicators at higher scales reinforces arguments in the literature that resilience enhancement should not rely solely on increasing facility quantity.

In addition, the station selection frequency analysis identifies a small number of high-probability hub nodes that consistently appear in optimal configurations across stochastic scenarios. This observation relates to prior studies on spatial heterogeneity and network criticality [3,43], which highlight the role of structurally important nodes in maintaining system functionality. By revealing a long-tailed distribution of station selection probabilities, the present study provides a probabilistic interpretation of network backbone formation under uncertainty. This offers practical insight into phased infrastructure investment, supporting a strategy that first secures high-probability core nodes before expanding toward scenario-dependent peripheral nodes.

The case study also reveals an important phenomenon: although the optimized layout increases the average coverage rate and average arrival time compliance rate to nearly 100%, the maximum arrival time compliance rate remains relatively low under random traffic resistance. This result is not a flaw in the model but a true reflection of real-world uncertainty. It clarifies two distinct types of emergency risks: first, the “spatial accessibility” risk caused by improper facility layout, which this study has systematically controlled to an extremely low level through global random optimization of node placement; and second, the “tail-end travel time” risk triggered by unpredictable extreme traffic congestion, which essentially exceeds the capabilities of static site selection models. This distinction holds significant managerial implications: the former can be fundamentally mitigated

through scientific site selection during the planning phase, while the latter requires dynamic strategies during the operational phase to address the issue, building upon the initial site selection.

Despite these contributions, several limitations should be acknowledged. First, for modeling tractability, pipeline demand points are approximated by road network nodes, and dispatch conflicts, resource limitations, and queuing effects are not explicitly modeled. Second, the study considers stochastic but relatively regular traffic variability and does not incorporate large-scale disruptive events such as extreme weather or major public incidents that may fundamentally alter network accessibility. Third, the improved iterative greedy search algorithm adopted in this study, while computationally efficient and effective in generating high-quality solutions, does not provide a theoretical guarantee of global optimality for the proposed CF-SRLP. This limitation stems from the problem's inherent complexity: the CF-SRLP is a generalization of the classic MCLP, which is NP-hard, and the integration of stochastic road resistance evaluated via Monte Carlo simulation further exacerbates the computational challenge, making the search for an exact global optimum in large-scale networks intractable. These simplifying assumptions and methodological choices may limit the model's applicability under extreme systemic disruptions or in contexts where proven optimality is required.

Future research should therefore explore more refined representations of pipeline–road network coupling and investigate integrated optimization of station location and dynamic routing under multiple sources of uncertainty. Building upon stochastic programming and robust planning approaches [7–9,22], future extensions may incorporate adaptive dispatch strategies and real-time traffic information to further reduce tail-end service delays. Additionally, integrating multi-objective formulations that explicitly trade-off cost, equity, and resilience could enhance decision-support capability for complex urban systems.

7. Conclusions

This study developed a candidate-free stochastic location optimization framework for gas emergency repair stations operating under uncertain traffic conditions. By eliminating reliance on predefined candidate sites and allowing all road network nodes to function as potential facility locations, the proposed model expands the feasible solution space and enables a truly global network-wide search. Stochastic road resistance is explicitly incorporated into travel time estimation, and a hybrid solution strategy combining an improved iterative greedy search with Monte Carlo simulation is designed to efficiently evaluate station configurations across a large number of scenarios. Application to the City H road network demonstrates stable and high coverage performance, achieving an average coverage rate of 97.77% across 20,000 stochastic scenarios. The results confirm a positive relationship between station scale and response performance, while revealing clear diminishing marginal returns beyond a critical threshold. Station selection frequency analysis further identifies structurally significant hub nodes that consistently form the backbone of effective configurations. From a methodological perspective, this study extends classical covering location models into a probabilistic, candidate-free framework. From a practical perspective, it provides quantitative decision-support for balancing coverage reliability, response timeliness, and resource efficiency in resilience-oriented planning of urban gas infrastructure systems under uncertainty.

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